

All n proof

hidden zeros and locality on the n -gon

Setting and result. B_n is a general rational ansatz with planar poles, assumed to vanish on every hidden zero of the n -gon. We prove that this hypothesis alone kills every coefficient except those of triangulations — crossings, underweight terms, and repeated poles all vanish — so that $B_n = c A_n$: locality and unitarity are outputs of the zeros, not inputs.

1 Zones and locked cuts

Label the vertices of a convex n -gon cyclically. A non-adjacent pair gives a chord $X_{i,j} = X_{j,i}$; an adjacent pair is a leg and is zero throughout. For any degree d write

$$B_{n,d} = \sum_{|M|=d} \frac{a_M}{\prod_{c \in M} X_c}, \quad B_n := B_{n,n-3},$$

where M is a multiset of chords. Its *support*, denoted $\text{supp } M$, is the set of distinct chords that occur in M , with their multiplicities forgotten. We call $d = n - 3$ *full weight* and $d < n - 3$ *underweight*; *squarefree* means that no chord is repeated. In the squarefree full-weight sector, *local* means that no two chords cross. A set of $n - 3$ pairwise non-crossing chords is exactly a triangulation. Thus such a denominator is non-local exactly when it contains two geometrically crossing chords. The short over v is $\sigma_v = X_{v-1,v+1}$; a chord crosses σ_v exactly when it ends at v .

The hidden zeros. A *zone* is a cyclic interval $T = \{w, \dots, w+k\}$ with $1 \leq k \leq n-4$. Write T^c for its complement. The notation $T | T^c$ records the partition of the vertices into the two sides T and T^c ; this partition is the *cut* determined by T . The two polygon legs $(w-1, w)$ and $(w+k, w+k+1)$ where the two sides meet are its *boundary legs*. The chords constrained by this cut are exactly those with one endpoint in T and the other in T^c ; none of them geometrically crosses either boundary leg. The hidden zero attached to T is the kinematic locus, written $\mathcal{H}_T$, on which every such chord is solved in terms of the chords along its own row and column,

$$X_{w+i,j} = X_{w,j} - X_{w,w+k+1} + X_{w+i,w+k+1} \quad (1 \leq i \leq k, j \notin T), \quad (1)$$

while chords with both ends in T , or both ends outside, stay free. The $\text{Tr } \phi^3$ amplitude vanishes identically on every such locus. Our standing hypothesis is that B_n does too — for every base w and every width k — and the goal is to show that this forces $a_M = 0$ for every non-triangulation M .

A vertex-based form of the hidden zero. Put $v = w + k + 1$ and define

$$A_r = X_{r,v} \quad (r \in T), \quad B_u = X_{w,u} - X_{w,v} \quad (u \in T^c). \quad (2)$$

We call these numbers the *vertex potentials*. For $r = w$, $A_w + B_u = X_{w,u}$ directly, while for $r = w + i$, equation (1) gives

$$X_{ru} = X_{w,u} - X_{w,v} + X_{r,v} = A_r + B_u.$$

Thus every cross-cut entry satisfies

$$X_{ru} = A_r + B_u \quad (r \in T, u \in T^c). \quad (3)$$

For the two adjacent cross-cut pairs, which are the boundary legs, (3) becomes

$$A_w + B_{w-1} = 0, \quad A_{w+k} + B_v = 0. \quad (4)$$

Conversely, choose arbitrary potentials satisfying (4), define every cross-cut chord by (3), ie one end point in T and the other in T^c , and leave all same-side chords free. For $r = w + i$ and $u = j$, the right-hand side of (1) becomes

$$(A_w + B_u) - (A_w + B_v) + (A_r + B_v) = A_r + B_u = X_{ru}.$$

Hence every equation in (1) holds. The hidden-zero locus is therefore exactly the set described by (3), (4), and freely chosen same-side chords.

Equivalently, its cross-cut entries obey the additive rectangle identities

$$X_{ru} + X_{r'u'} = X_{ru'} + X_{r'u} \quad (r, r' \in T, u, u' \in T^c). \quad (5)$$

Equation (3) immediately implies (5). Conversely, taking $r' = w$ and $u' = v$ in (5) gives

$$X_{ru} = X_{w,u} - X_{w,v} + X_{r,v},$$

which recovers (2) and (3). Thus the potential description is precisely the classical Born-free row-and-column parametrization written without privileging a particular basis.

The potentials have the gauge freedom

$$(A, B) \mapsto (A + s, B - s),$$

which leaves every chord coordinate $A_r + B_u$ unchanged. The formulas (2) choose the convenient gauge $B_v = 0$. The cross-cut sector consequently has

$$n \text{ potentials} - 1 \text{ gauge freedom} - 2 \text{ boundary equations} = n - 3$$

independent coordinates. In the curve below we may use a different gauge representative, because only the gauge-invariant sums $A_r + B_u$ define the chord coordinates.

In the language of discrete connections: setting $f = -A$ on T and $f = B$ on T^c makes every cross-cut entry the exact 1-cochain $X_{ru} = f(u) - f(r)$ on the complete bipartite graph, (5) is zero holonomy around a four-cycle, and since four-cycles generate the cycle space, the hidden zero says precisely that this connection is flat — with $(A, B) \mapsto (A + s, B - s)$ the shift of the potential f by a constant.

The linkage graph. For a chord set S and zone T in the n -gon, the *linkage graph* $\Gamma(S, T)$ has vertices $1, \dots, n$ and edges called *links*. Its two *boundary links* join $w - 1$ to w and $w + k$ to $w + k + 1$; these represent the two boundary legs. For every chord $X_{ru} \in S$ with $r \in T$ and $u \in T^c$, draw a *chord link* from r to u . A boundary link represents $A_r + B_u = 0$, while a chord link represents $X_{ru} = A_r + B_u$.

The connected component containing a vertex consists of that vertex and every vertex reachable from it by a path of links. An *isolated vertex* touches no link, even if chords of S lying entirely on one side of the cut touch it in the polygon. The links impose no relation between potentials in different components.

The cut is *unlocked* for S when every component is a *star*: all its links share one vertex; an isolated vertex also counts as a star. Because every link joins the two sides, a component is not a star exactly when it has at least two vertices in T and at least two in T^c . Call S *fully locked* when no cut is unlocked. Figure 1 shows the dictionary.

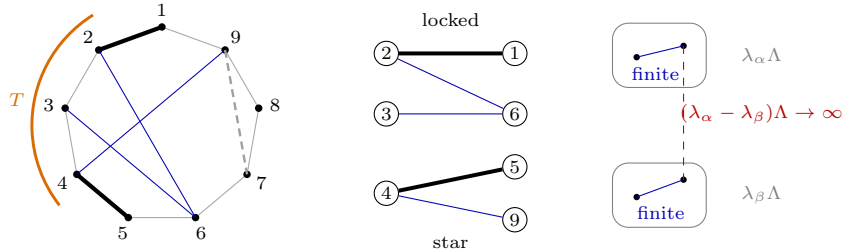


Figure 1: Left: the zone $T = \{2, 3, 4\}$ on the 9-gon, with boundary legs $(1, 2), (4, 5)$ bold and the three cross-cut chords of S , $X_{2,6}, X_{3,6}, X_{4,9}$, in blue. The dashed gray chord $X_{9,7}$ also belongs to S . Middle: the corresponding links in the linkage graph $\Gamma(S, T)$. One component contains $\{2, 3\}$ on the T side and $\{1, 6\}$ on the T^c side, so it is locked; the component at 4 is a star. Since $7, 9 \in T^c$, $X_{9,7}$ creates no link; vertex 7 is therefore isolated in the linkage graph even though a chord of S ends there. Vertex 8 is also isolated. By contrast, $X_{2,9}$ would be drawn because $2 \in T$ and $9 \in T^c$. It adds a link between the already existing vertices 2 and 9; no vertex is duplicated. This new link joins the two connected components shown — the upper locked component and the lower star component — into one connected component, which is not a star. One locked component suffices: this cut is not unlocked. Right: the scaling. Each component C_α is given a distinct label $\lambda_\alpha, \lambda_\beta$, etc., and every vertex inherits its component's label. A chord whose two endpoints lie in the *same* component (blue) has its Λ terms cancel and stays *finite*; a chord joining *different* components (red, dashed) keeps a term $(\lambda_\alpha - \lambda_\beta)\Lambda$ and blows up as $\Lambda \rightarrow \infty$. Only the difference of labels matters, not their size.

Why an unlocked cut kills a coefficient. Fix the multiset M whose coefficient we test, let $S = \text{supp } M$, and suppose some cut $T | T^c$ is unlocked for S : every component of $\Gamma(S, T)$ is a star. The linkage graph is built once, from M alone; the curve constructed from it below is then substituted into *every* term of B_n , not only the term indexed by M .

The idea: we lay a curve inside the hidden-zero locus along which every chord of M stays finite, while a chord crossing the cut blows up unless it is a link of $\Gamma(S, T)$. In the limit, the identity $B_n = 0$ collapses to a sum of Laurent monomials in independent variables, one per surviving term — and M 's monomial is among them, so $a_M = 0$.

The curve. Label the components $C_1, \dots, C_q$. In each component containing a link, choose a vertex through which every link passes and call it the *centre* (for a single link, either endpoint will do); every other vertex of that component is a *leaf* and meets exactly one link. Call a leaf a *chord leaf* when its link is a chord link. Choose distinct real numbers $\lambda_1, \dots, \lambda_q$, one per component, and set, for $r \in T \cap C_\alpha$ and $u \in T^c \cap C_\alpha$,

$$A_r = \lambda_\alpha \Lambda + a_r, \quad B_u = -\lambda_\alpha \Lambda + b_u,$$

where the offset a_r (resp. b_u) is an independent nonzero variable when the vertex is a chord leaf, and 0 otherwise — so centres, leaves of boundary links, and isolated vertices all carry offset zero. Define every cross-cut chord by the split (3), and give every chord with both endpoints in T , or both in T^c , its own independent nonzero variable: (3) concerns only cross-cut pairs and places no condition on same-side chords. For each fixed choice of these finite parameters, varying Λ gives a one-parameter curve. Allowing the finite parameters themselves to vary gives the family of curves used below to separate the surviving denominators.

The curve lies in the zero locus. By the vertex-potential characterization above, it is enough to verify the two boundary-leg equations. The endpoints of a boundary link belong to the same component, so their Λ terms cancel. Their offsets also vanish: one endpoint is the centre, while the other is a leaf whose unique link is a boundary link rather than a chord link. Hence

$$A_w + B_{w-1} = a_w + b_{w-1} = 0, \quad A_{w+k} + B_v = a_{w+k} + b_v = 0$$

for every Λ . Therefore, for every fixed choice of the finite parameters, the potentials define through (3) a curve lying entirely in the hidden-zero locus. Along every such curve the standing hypothesis gives $B_n = 0$ identically in Λ .

Values along the curve. (Figure 1, right). A cross-cut chord with $r \in T \cap C_\alpha$ and $u \in T^c \cap C_\beta$ takes the value

$$X_{ru} = (\lambda_\alpha - \lambda_\beta)\Lambda + a_r + b_u.$$

If $\alpha \neq \beta$, it grows like Λ , and every term of B_n containing it dies as $\Lambda \rightarrow \infty$. If $\alpha = \beta$, the Λ terms cancel and the chord stays finite; moreover, since all links of a star pass through its centre, the component has exactly one vertex on one side of the cut, so the pair r, u must consist of the centre and a leaf, and the chord's value is that leaf's offset. A finite cross-cut chord is therefore precisely a chord link of $\Gamma(S, T)$, hence a chord of $\text{supp } M$, and its value is the variable attached to its chord leaf. In particular every chord of M survives: its cross-cut members are links by construction, and its same-side members keep the variables assigned to them above. Terms other than M may survive as well; the independent variables are what now separates them.

Independence of the finite coordinates. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Let $\mathcal{L}$ be the set of chord leaves of $\Gamma(\text{supp } M, T)$, and let $\mathcal{D}$ be the set of same-side chords. For each $\ell \in \mathcal{L}$, let $c(\ell)$ denote the centre joined to ℓ .

Consider the coordinate projection

$$\pi : \mathcal{H}_T \longrightarrow \mathbb{K}^{|\mathcal{L}|+|\mathcal{D}|}, \quad \pi(x) = \left((X_{c(\ell), \ell}(x))_{\ell \in \mathcal{L}}, (X_d(x))_{d \in \mathcal{D}} \right).$$

Thus π simply forgets every coordinate of the point $x \in \mathcal{H}_T$ except the chord-leaf coordinates and the same-side chord coordinates.

We now construct a right inverse of π . Begin with completely arbitrary values $(y_\ell)_{\ell \in \mathcal{L}}, (z_d)_{d \in \mathcal{D}}$. Fix $\Lambda = 0$ in the construction above, use y_ℓ as the offset of the chord leaf ℓ , and set $X_d = z_d$ for every same-side chord $d \in \mathcal{D}$. Keep the offsets of centres, boundary leaves, and isolated vertices equal to zero, as before. The boundary-leg calculation already proved that this assignment defines a point of $\mathcal{H}_T$. Denote that point by $s(y, z) \in \mathcal{H}_T$. If $\ell \in T$, its centre lies in T^c , and therefore $X_{\ell, c(\ell)} = A_\ell + B_{c(\ell)} = y_\ell + 0 = y_\ell$. Combining the two possible positions of a chord leaf with the assignment of the same-side coordinates gives (read first equation backwards if it helps)

$$y_\ell = \begin{cases} X_{\ell, c(\ell)}(s(y, z)), & \ell \in T, \\ X_{c(\ell), \ell}(s(y, z)), & \ell \in T^c, \end{cases} \quad z_d = X_d(s(y, z)) \quad \implies \quad \pi(s(y, z)) = (y, z), \quad \pi \circ s = \text{id}. \quad (6)$$

Here the two orders of the endpoints merely keep the endpoint in T first; the chord variable itself is unoriented, so $X_{ij} = X_{ji}$. Thus s is a right inverse of π , or a section of π . Geometrically, for every possible tuple (y, z) , the section selects a point of the hidden-zero locus having exactly those selected coordinate values (geometrically, the hidden-zero locus contains a copy of the entire (y, z) -coordinate space).

Now suppose that a polynomial P in these selected coordinates vanishes everywhere on $\mathcal{H}_T$. Thus, $P \circ \pi = 0$ on $\mathcal{H}_T$. For every arbitrary choice of (y, z) , the constructed point $s(y, z)$ lies in $\mathcal{H}_T$. Therefore

$$0 = P(\pi(s(y, z))) = P\left((X_{c(\ell), \ell}(s(y, z)))_{\ell \in \mathcal{L}}, (X_d(s(y, z)))_{d \in \mathcal{D}}\right) = P((y_\ell)_{\ell \in \mathcal{L}}, (z_d)_{d \in \mathcal{D}}).$$

The second line is the definition of the projection π , and the last equality follows from (6). Thus P vanishes for every possible value of all its variables. A polynomial over $\mathbb{R}$ or $\mathbb{C}$ with this property is the zero polynomial. Hence $P = 0$, and the selected coordinates are algebraically independent.

Separation. The variables above are one common family of coordinates used to evaluate every surviving term. Every surviving cross-cut chord receives the variable of its unique chord leaf, while every surviving same-side chord receives its own variable. Hence distinct surviving chords give distinct Laurent monomials, including multisets that differ only in multiplicity. We choose the finite variables generically and nonzero, avoiding the finitely many exceptional values at which a denominator vanishes. If a linear relation among the resulting Laurent monomials existed, multiplying the entire relation by one common monomial would clear every negative exponent and produce a polynomial relation among the same independent variables. The independence just proved forces that polynomial relation, and hence the original Laurent relation, to be trivial.

Taking $\Lambda \rightarrow \infty$ in the identity $B_n = 0$ therefore leaves a sum of distinct Laurent monomials whose coefficients must vanish separately. The multiset M survives, so among those coefficients is a_M :

$$\boxed{\text{some cut unlocked for } \text{supp } M \implies a_M = 0.}$$

A final gauge observation. The component labels used in the killing limit are themselves defined only up to a common shift. Indeed, replacing

$$\lambda_\alpha \longmapsto \lambda_\alpha + c \quad \text{for every component } C_\alpha$$

changes the potentials pointwise along the curve by

$$A_r \longmapsto A_r + c\Lambda, \quad B_u \longmapsto B_u - c\Lambda.$$

This is precisely the gauge transformation $(A, B) \mapsto (A+s, B-s)$ with $s = c\Lambda$, so every chord coordinate $X_{ru} = A_r + B_u$ remains unchanged. Consequently the killing limit depends only on the relative difference $\lambda_\alpha - \lambda_\beta$. For q components, its component-scaling directions therefore form

$$\mathbb{R}^q / \langle (1, \dots, 1) \rangle,$$

with $q - 1$ independent relative directions. The absolute scale assigned to a component is gauge; only the scaling between different components enters the kinematics.

Moreover, the labels λ_α form a 0-cochain on the graph of components, a cross-component chord sees its gradient $(d\lambda)_{\alpha\beta}$, and the common mode is the kernel of d .

2 From locked cuts to locality

The zone statement. The remaining content is entirely geometric — one statement about chords on the n -gon, with no mention of the ansatz:

$$\text{If } |S| \leq n - 3 \text{ and } S \text{ is not a triangulation, then some cut is unlocked for } S.$$

We prove this in Section 5. Here we show it delivers locality. Granting it, the kill criterion of Section 1 does the rest: an unlocked cut forces $a_M = 0$, and every non-triangulation denominator meets the hypothesis.

Locality, unitarity, and the underweight and repeated sectors. A full-weight multiset with a repeated chord has $|\text{supp } M| \leq n - 4$, not a triangulation; an underweight multiset of degree $d \leq n - 4$ has $|\text{supp } M| \leq d \leq n - 4$, not a triangulation; a squarefree full-weight non-local multiset has $|\text{supp } M| = n - 3$ with a crossing, again not a triangulation. In every case the zone statement gives an unlocked cut and the kill criterion gives $a_M = 0$. What survives is exactly the squarefree non-crossing full-weight sector — the triangulations, which are fully locked, as they must be: the amplitude vanishes on every hidden zero, so no triangulation coefficient can be forced to zero. Rodina's uniqueness theorem then equates all of them, $B_n = cA_n$, and unitarity follows.

It remains to prove the zone statement.

3 The reduction to terminal sets.

Let $\deg_S(v)$ count the chords of S at v . Call v *free* when $\deg_S(v) = 0$, and S *terminal* when no vertex is free. The plan: a free vertex either hands us an unlocked cut outright or lets the polygon contract around it, so the problem strictly shrinks until it is terminal — and fully locked terminal sets will turn out to need at least $n - 2$ chords, one more than any denominator can afford.

The contraction and lift machinery below applies to any chord set on the n -gon. Two of its consumers track distinguished chords through the cascade: the terminal bound of Section 4 tracks a *crossed short* σ_v together with a crosser $X_{v,u} \in S$, and the two-crossing-chords lemma of Section 5 tracks a crossing pair. Each introduces its own tracked chords, and the claims below record how contraction preserves them whenever they are being tracked.

The two-vertex cuts. For $T = \{w, w + 1\}$, the chords used in $\Gamma(S, T)$ have exactly one endpoint in $\{w, w + 1\}$, and the linkage graph fails to be a star forest exactly when S contains

$$\underbrace{X_{w,w+2}}_{\text{the special}}, \quad \underbrace{X_{w+1,w-1}}_{\text{the bare}}, \quad \text{or} \quad \underbrace{\{X_{w,j}, X_{w+1,j}\}}_{\text{a column pair at apex } j} \quad (\text{R})$$

(Figure 2). Indeed the special and the bare each join the two boundary legs into a path of three links. A column pair joins the two boundary legs through its apex, giving a path of four links; either boundary leg together with the two column chords already forms a path of three links. Thus each case has a connected component that is not a star. Conversely, chords at w together with the leg $(w - 1, w)$ form a star at w , chords at $w + 1$ with the other leg a star at $w + 1$, and without (R) nothing merges the two stars. We call (R) the two-vertex rule.

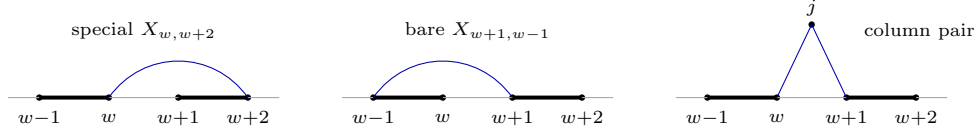


Figure 2. The two-vertex rule (R): the only three ways to lock the cut $\{w, w + 1\}$. Its two boundary legs $(w - 1, w)$ and $(w + 1, w + 2)$ are bold; the internal polygon leg $(w, w + 1)$ is gray and is not a link of the linkage graph. The special and the bare are the shorts over $w + 1$ and over w .

Bare free tip. If p is free and $\sigma_p \notin S$, the cut $\{p - 1, p\}$ is unlocked: every chord of S with one endpoint in $\{p - 1, p\}$ ends at $p - 1$. Hence the component containing the boundary leg $(p - 2, p - 1)$ is a star centred at $p - 1$, while the other boundary leg $(p, p + 1)$ is an isolated one-link component.

Contraction. Let p be free and suppose $\sigma_p \in S$. Let φ delete p and relabel the remaining vertices cyclically, and define the descended chord set

$$S' = \varphi(S \setminus \{\sigma_p\}).$$

Because p is free, no chord of S ends at p . The only chord whose endpoints become adjacent after p is deleted is $\sigma_p = X_{p-1,p+1}$, and deletion is injective on all remaining vertices. Thus no other chord degenerates or is identified with another, and

$$|S'| = |S| - 1.$$

Contraction also preserves the distinguished configurations used below. First suppose that a crossed short

$$\sigma_v, X_{v,u} \in S, \quad X_{v,u} \text{ crossing } \sigma_v,$$

is being tracked. Since p is free, it is none of $v - 1, v, v + 1, u$. Hence deleting p leaves $v - 1, v, v + 1$ consecutive, so $\varphi(\sigma_v)$ remains a short. If $\varphi(X_{v,u})$ became a leg, then $X_{v,u}$ would have to be the short σ_p . But σ_v crosses $X_{v,u}$, whereas a chord crosses σ_p exactly when it ends at p ; this would make p an endpoint of σ_v , contrary to p being free. Therefore $\varphi(X_{v,u})$ also remains a chord. Deleting a vertex distinct from the four endpoints preserves their cyclic order, so $\varphi(X_{v,u})$ still crosses $\varphi(\sigma_v)$.

The same reasoning preserves any distinguished crossing pair: a free vertex is not one of its four endpoints, while σ_p is uncrossed because a chord crossing it would have to end at p . Hence neither member of the pair is removed, and their cyclically alternating endpoints remain alternating downstairs.

In the comparison below, the original n -gon is called *upstairs* and the contracted $(n - 1)$ -gon *downstairs*.

Unlocked cuts lift. Here φ acts on the chord diagram alone: it deletes p (equivalently, sends $\sigma_p \rightarrow 0$, so that σ_p becomes the leg $(p-1, p+1)$) and asks which descended diagrams remain fully locked. No Laurent coefficient of B_n is extracted — this is a geometric statement about locked diagrams, not a residue of the ansatz.

We show every unlocked cut of S' lifts to an unlocked cut of S . Given a cut T' unlocked for S' , reinsert p and check that the resulting upstairs cut is unlocked. Throughout, “ T' unlocked for S' ” means every component of $\Gamma(S', T')$ is a star — the fact we lean on at each step. Let the *scar* be the downstairs leg ab , with $a = \varphi(p-1)$ and $b = \varphi(p+1)$; reinserting p always places it between a and b . Later we will use the contrapositive: if S is fully locked (no cut unlocked), so is S' .

Scar interior to a side. If ab is not a boundary leg of T' , then a, b lie on one side, and there is a single lift: p joins them there. Then $\sigma_p = (a, b)$ has both endpoints on that side, so it contributes no link, and every other chord crosses the lift exactly when its image crosses T' . Hence $\Gamma(S, \text{lift})$ is $\Gamma(S', T')$ with the isolated vertex p adjoined — still all stars. The lift remains unlocked.

Scar a boundary leg. If ab is a boundary leg of T' , then a, b lie on opposite sides, and there are two lifts: reinserting p makes the new boundary leg either (a, p) or (p, b) . Here ab is a boundary link of $\Gamma(S', T')$, so, since T' is unlocked, the component containing it is a star; say its centre is a , the mirror case being identical. Then b is a leaf: no other link of that component ends at b . Take the lift that places p on b 's side (the leaf's side), making the upstairs boundary leg $(p-1, p) = (a, p)$. Now $\sigma_p = (a, b) \in S$ becomes a cross-cut chord, contributing the link (a, b) ; together with the new boundary link (a, p) and the old star's links, every link of the component passes through centre a , while b and p are leaves. So the component is still a star, no other component changed, and the lift remains unlocked.

Terminal sets, and what remains. The machinery above shrinks a fully locked set free vertex by free vertex, preserving full lock, until it is terminal. One remark for later: in a terminal set every vertex carries a chord, so the tip of any short carries one, and a chord at the tip crosses the short — *in a terminal set every short is crossed*. Everything now rests on one quantitative fact, proved in Section 4 and consumed in Section 5:

$$\text{A fully locked terminal set has } |S| \geq n - 2.$$

Two counting facts about a fully locked terminal S with $|S| \leq n - 3$:

(a) *Shorts exist.* Suppose S had no shorts. Then at each of the n two-vertex cuts $\{w, w+1\}$, rule (R) has no special and no bare on offer — both are shorts — so it demands a column pair $\{X_{w,j}, X_{w+1,j}\}$: two chords at a common apex j whose feet are the adjacent pair $w, w+1$. Say the apex j serves the polygon leg $(w, w+1)$. How many legs can one apex serve? Fix j and let $d_j = \deg_S(j)$. The feet of j 's chords are d_j distinct vertices, and j serves the leg $(w, w+1)$ exactly when both w and $w+1$ are among them. Around the polygon the feet form disjoint consecutive runs — never the full circle, since j itself is not a foot — and a run of ℓ vertices contains $\ell - 1$ adjacent pairs; summing over the runs, d_j feet contain at most $d_j - 1$ adjacent pairs. So the apex j serves at most $d_j - 1$ legs. Every one of the n legs is served by some apex, so counting the pairs (leg, apex serving it):

$$n \leq \sum_{j=1}^n (d_j - 1) = 2|S| - n,$$

using $\sum_j d_j = 2|S|$ (each chord has two endpoints) and the fact that the sum runs over all n vertices — terminality makes every term nonnegative, so no vertex contributes a hidden -1 . Hence $|S| \geq n$: too big for $|S| \leq n - 3$. (This is the same count that forces shorts throughout this subject: a chord pays for at most one leg beyond its apex's first, and only a short escapes the ledger, covering two legs by itself — as the special of one cut and the bare of the next.)

(b) *At least six vertices have degree one.* Terminality makes every term of $\sum_v (\deg_S(v) - 1) = 2|S| - n \leq n - 6$ nonnegative, so at most $n - 6$ vertices have degree ≥ 2 .

4 Leaf smoothing and the terminal bound.

Leaf smoothing. The engine is that a degree-one vertex can be *smoothed away*: deleted from the polygon so that full lock survives while the chord count drops. The intuition: a leaf's chord is never load-bearing on its own — full lock at the leaf's two 1-zones forces a *mate* right next to it, and the mate takes over every job the leaf's chord was doing.

Lemma 1 (Smoothing) *Let S be fully locked on the n -gon, $n \geq 6$, let $\deg_S(p) = 1$ (so p is a leaf) with sole chord $e = X_{p,u}$, put $a = p-1$, $b = p+1$, and let φ delete p .*

(1) If $\sigma_p \notin S$: $S_0 = \varphi(S \setminus \{e\})$ is fully locked on the $(n-1)$ -gon.

(2) If $\sigma_p \in S$: of the two slides (Figure 3)

$$S_a = \varphi(S \setminus \{e, \sigma_p\}) \cup \{X_{\varphi(a), \varphi(u)}\}, \quad S_b = \varphi(S \setminus \{e, \sigma_p\}) \cup \{X_{\varphi(b), \varphi(u)}\}$$

(omit the added chord if it degenerates to a leg; identify it if already present), at least one is fully locked on the $(n-1)$ -gon.

Either way the chord count drops by at least one.

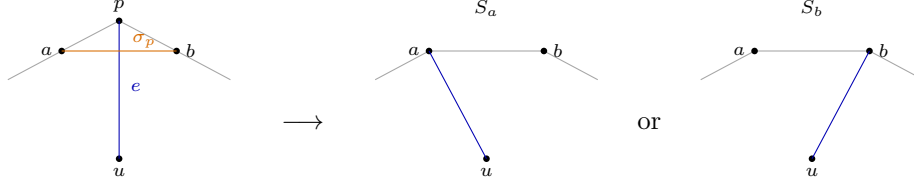


Figure 3. Leaf smoothing in case (2): the degree-one vertex p is deleted, the endpoints of σ_p become adjacent along the new boundary (the *scar*), and e slides to one of the two scar endpoints. At least one slide is fully locked.

Proof. A downstairs cut has one lift if the scar ab is not a boundary leg, and two lifts if it is. Every lift is a legal cut: a downstairs cut has at least two vertices on each side, and reinserting p only enlarges one side. Every chord avoiding p keeps the same two sides. Thus only pu , σ_p , and, when the scar is a boundary leg, the replacement of that boundary leg can change the linkage graph.

First suppose $\sigma_p \notin S$. Since $\deg_S(p) = 1$, the sole chord at p is $e = X_{p,u}$, so u is the only apex p can reach. Rule (R) at $\{a, p\}$ (special $\sigma_p \notin S$) forces au to close a triangle: either $u = p - 2$, making au a leg, or $X_{a,u} \in S$. Symmetrically at $\{p, b\}$: $u = p + 2$, or $X_{b,u} \in S$. Either way the pairs au and bu are both available: as chords of S_0 , or — for at most one of them — as a polygon leg. So for each $z \in \{a, b\}$, every downstairs cut separating z from u carries a zu link.

Suppose for a contradiction that a downstairs cut D were unlocked for S_0 . If the scar is not a boundary leg, then a, b lie on the same side and the lift is unique. If u also lies on the same side as p , pu creates no link, so the lift is the same star forest plus the isolated vertex p . If u lies opposite, both au and bu are links of $\Gamma(S_0, D)$. They meet only at u , so their component, being a star, is centred at u ; adding the link pu in the lift merely adds another leaf to that star. Thus the lift is unlocked in either case, contrary to full lock of S .

If the scar is a boundary leg of D , put p on the same side as u . Then pu creates no link and σ_p is absent. The only change is that the downstairs boundary link ab is replaced by pb if u lies with a , or by ap if u lies with b . In $\Gamma(S_0, D)$ the boundary link ab lies in a star (since D is unlocked for S_0). The lift replaces ab by pb . If the star's centre is b , then a is a leaf, and swapping the leaf a for p leaves a star centred at b . If the centre is a , then b is a leaf, and pb splits the component into the star at a (now missing the leaf b) and the one-link star pb . Every other component is unchanged. Hence this lift is also unlocked, again a contradiction. Therefore S_0 is fully locked. This proves (1).

Now suppose $\sigma_p \in S$. Put

$$R = \varphi(S \setminus \{pu, \sigma_p\}), \quad S_z = R \cup \{X_{z,u}\} \quad (z \in \{a, b\}),$$

say $S_z = R$ if $X_{z,u} = 0$, and $\{z, \bar{z}\} = \{a, b\}$. Call a downstairs cut *bad for the z -slide* if it is unlocked for S_z . Every lift of a bad cut is locked for S , because S is fully locked.

We first determine exactly what a bad cut can look like. Two elementary facts about star forests will be used. Replacing one leaf of a star by a new leaf preserves a star forest. attaching a link from v to the new isolated vertex p creates a non-star exactly when v was a leaf of a star having at least two links; attaching at a centre, an isolated vertex, or a one-link component leaves a star forest.

If the scar is not a boundary leg, a, b, p lie on the same side of the unique lift. We can assume the vertex u lies opposite: otherwise lifting only adds the isolated vertex p and we are done. If the slide chord zu is not already in R (and not a leg), lifting would merely replace it by pu and would preserve the star forest. Thus a bad cut must already have a zu link independent of the slide. There are exactly two possibilities:

- (I_c) $X_{z,u}$ is a chord link. The component is a star centred at z with another link zy , which may be a chord link or a boundary link;
- (I_l) $X_{z,u}$ is a boundary link. The component is a star centred at z with another link zy , necessarily a chord link, since zu is boundary.

Indeed the lift retains this independent zu link and adds pu , so the two star-forest facts give precisely the stated extra link.

If the scar is a boundary leg and u lies with $\bar{z}$, use the lift putting p on that side. Downstairs the distinct links $z\bar{z}$ and zu share z , so their component, being a star, is centred at z . The lift drops the slide chord $X_{z,u}$ if $X_{z,u} \notin R$; the surviving links $(z, \bar{z})$ and (z, p) both pass through z , so it is still a star forest. Hence a bad cut must put u with z . Use the lift putting p on that side. Downstairs the scar component contains $z\bar{z}$; upstairs it retains that link and attaches the new link $p\bar{z}$. By the elementary facts of star forests, this can be a non-star only in the third possibility:

- (II) $X_{z,\bar{z}}$ is a boundary link. The component is a star centered at z with another link zy , necessarily a chord link, and $y \neq \bar{z}$.

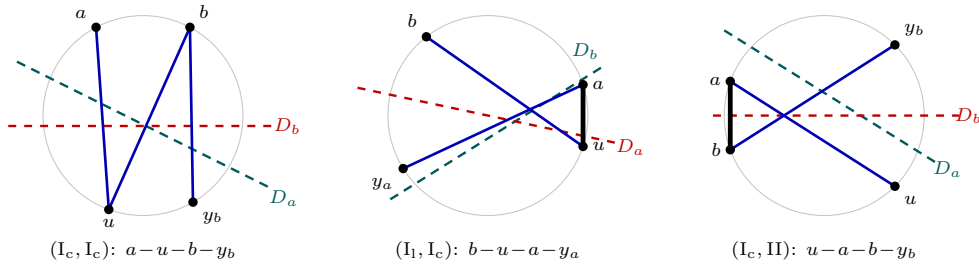
$X_{y,z}$ cannot be the other boundary leg: both boundary legs would then end at z , making z 's side the singleton $\{z\}$ — but $u \neq z$ lies on that side. These three types — I_c , I_l , II — are exhaustive. Each bad z -cut separates $\bar{z}$ from u .

It remains to show that bad cuts for the two slides cannot coexist. Let D_a, D_b be bad for the a - and b -slides, respectively, and denote their extra far endpoints by y_a, y_b . Some type pairings are impossible at once. The pair (I_l, I_l) would make u adjacent to both a and b , impossible on the downstairs polygon, which has at least five vertices. If D_a has type I_l and D_b type II , then u is a 's neighbour other than b , while D_b separates a from both of its neighbours b, u ; the side containing a would be the illegal singleton $\{a\}$. The pairing (II, I_l) is its mirror.

For every remaining pairing except (II, II) , one of the two allegedly unlocked linkage graphs contains the following *three-link-path*:

type(D_a)	type(D_b)	graph tested	path	
I_c	I_c or I_l	$\Gamma(S_b, D_b)$	$a - u - b - y_b$	(T)
I_l	I_c	$\Gamma(S_a, D_a)$	$b - u - a - y_a$	
I_c	II	$\Gamma(S_b, D_b)$	$u - a - b - y_b$	
II	I_c	$\Gamma(S_a, D_a)$	$u - b - a - y_a$	

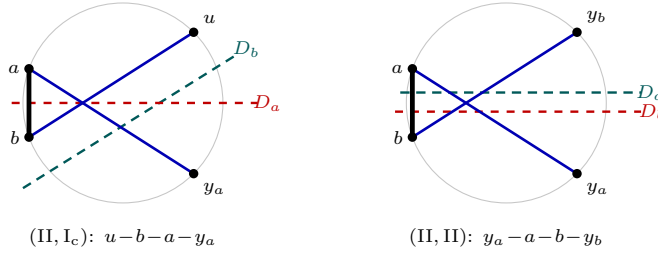
Every bad cut is one of the three types I_c, I_l, II , and the last paragraph already excludes (I_l, I_l) , (I_l, II) , and (II, I_l) ; the six surviving pairings of a bad a -cut D_a with a bad b -cut D_b are the ones drawn below (skipping the symmetrical (I_c, I_l) case as in the table above). In each panel we draw *both* cuts on the same polygon — the tested cut D_z dashed red, the other dashed teal — and exhibit three links forming a path with two vertices on each side of D_z : a locked component in a graph the badness of D_z claims is a star forest, the contradiction. The picture rests on the same three moves. There is one far vertex u , the far endpoint of the single leaf chord $e = X_{p,u}$, shared by both slides. A bad cut of any type at a vertex z carries, by its definition above, *two* links at z : the connector $X_{z,u}$ (or the scar $X_{z,\bar{z}}$) and a second link $X_{z,y}$. And every bad cut separates $\bar{z}$ from u , so a chord pinned at one scar endpoint by one cut is forced to *cross* the other — the transfer that glues the two cuts' link-pairs into one path. Reading each panel by its two types:



(I_c, I_c) , test D_b : D_a is I_c , so $X_{a,u}$ is a chord. Since D_b separates a from u , it crosses D_b , so $X_{a,u} \in \Gamma(S_b, D_b)$. D_b is I_c : its links $X_{b,u}$ and X_{b,y_b} complete $a - u - b - y_b$.

(I_l, I_c) , test D_a : D_b is I_c , so $X_{b,u}$ is a chord, crossing D_a (which separates b from u), so $X_{b,u} \in \Gamma(S_a, D_a)$. D_a is I_l : its links — the boundary leg (a, u) and the chord X_{a,y_a} — complete $b - u - a - y_a$.

(I_c, II) , test D_b : D_a is I_c , so $X_{a,u}$ is a chord, crossing D_b , so $X_{a,u} \in \Gamma(S_b, D_b)$. D_b is II : its links — the scar (a, b) and the chord X_{b,y_b} — complete $u - a - b - y_b$.



(II, I_c), test D_a : mirror of the previous — D_b is I_c, so $X_{b,u}$ crosses D_a ; D_a is II, with links the scar (a, b) and X_{a,y_a} , completing $u-b-a-y_a$.

(II, II), test D_b : both scars are the one boundary leg (a, b) , so the two b -sides nest; u on a 's side of D_a but b 's side of D_b puts y_a on D_b 's b -side, so the chord X_{a,y_a} crosses D_b . With the scar (a, b) and D_b 's chord X_{b,y_b} , the path is $y_a-a-b-y_b$.

In each panel the first and third path-vertices lie on one side of the tested cut, the second and fourth on the other: a locked component in a graph whose badness declares it a star forest. With (I₁, I₁), (I₁, II), (II, I₁) already excluded, no bad a -cut and bad b -cut coexist; at least one slide is fully locked, and leaf smoothing is proved. $\square$

The terminal bound. A fully locked terminal set has $|S| \geq n - 2$. Suppose not; take a counterexample — terminal, fully locked, $|S| \leq n - 3$ — with n minimal. Since S covers each vertex, $2|S| \geq n$; hence $n \leq 2|S| \leq 2n - 6$, so necessarily $n \geq 6$ and leaf smoothing is available. By (a) it contains a short σ_v , crossed (terminality) by some $X_{v,u}$. By (b) at least six vertices have degree one. Only four vertices are *unsafe* to delete while preserving the pair $(\sigma_v, X_{v,u})$: $v - 1, v, v + 1$, and u . Pick a degree-one vertex p not equal to these four and smooth it: the result S' is fully locked on the $(n - 1)$ -gon with $|S'| \leq n - 4$.

Smoothing may create free vertices. While one exists, say q : full lock forbids the bare-tip configuration, so $\sigma_q \in S'$; contract q . Full lock persists by the lift claim, the crossed pair persists as in the contraction claim, and vertices and chords drop by one each, so $|S'| \leq n' - 3$ is maintained. If the cascade stops at a terminal set on $n' \geq 6$ vertices, that is a smaller counterexample — contradicting minimality. And it cannot reach $n' = 5$: a non local 5-gon has a bare free tip 5, i.e. an unlocked cut, contradicting the full lock that the cascade preserves. So no counterexample exists. $\square$

(The tracking of $(\sigma_v, X_{v,u})$ through the cascade is not decorative: the 5-gon *does* carry a fully locked two-chord set, the apex pair $\{X_{1,3}, X_{1,4}\}$ — each chord is the special of one cut and the bare of another, and together they are a column pair for the fifth — but it contains no crossed short, so a cascade carrying the pair can never land on it.)

5 Proof of the zone statement.

Two crossing chords unlock a cut. *If $|S| \leq n - 3$ and S contains two crossing chords, then some cut is unlocked for S .*

Proof. Suppose every cut were locked. While a free vertex q exists, it is no bare free tip (an unlocked cut), so $\sigma_q \in S$. Contract q as in Section 3. Since σ_q is uncrossed, neither chord of the pair is σ_q ; the pair's four endpoints carry chords, so q is none of them; and neither image degenerates to a leg, since a chord becomes a leg only when it is the short over the deleted vertex — that short is σ_q itself. The crossing pair survives. The descent cannot reach the 5-gon, because no nonlocal in the 5-gon is fully locked. So the cascade is forced to halt at a *terminal* stage with $n' \geq 6$, still fully locked, still with $|S'| \leq n' - 3$ — contradicting the terminal bound. $\square$

Small supports unlock a cut. *If $|S| \leq n - 4$, then some cut is unlocked for S .*

Proof. Suppose S were fully locked and contract as above; each step lowers $|S|$ and n by one, so the deficit $|S| \leq n - 4$ persists, and if the chord count hits zero every cut is unlocked outright (two boundary links, two stars). The descent cannot reach the 5-gon: there $|S'| \leq 1$, and a lone short σ_v locks only the cuts $\{v - 1, v\}$ and $\{v, v + 1\}$, leaving $\{v + 1, v + 2\}$ with no special, no bare, and no column pair — unlocked by rule (R). So the cascade halts at a terminal stage with $n' \geq 6$; but a terminal set has $n' \leq 2|S'| \leq 2(n' - 4)$, and the terminal bound demands $|S'| \geq n' - 2$ against $|S'| \leq n' - 4$. $\square$

The zone statement follows: let $|S| \leq n - 3$ with S not a triangulation. If $|S| = n - 3$, then S contains two crossing chords — a set of $n - 3$ pairwise non-crossing chords *is* a triangulation — and the two-crossing-chords lemma gives an unlocked cut. If $|S| \leq n - 4$, the small-supports lemma does. With Section 2, every non-triangulation coefficient vanishes for every n — underweight and repeated denominators included — and $B_n = c A_n$.